Sri Lanka Institute of Information Technology

Faculty of Computing, Department of Computer Systems Engineering

IE3122: Network Security

Week 6 Tutorial Worksheet: Layer 4 Attacks and Defenses

Field	Detail
Based on	Week 6, Layer 4 Attacks and Defenses lecture (Layer 4 fundamentals, sockets, TCP handshake and teardown, six TCP attacks, UDP fundamentals, four UDP attacks)
Learning Outcomes	LO1 (describe network vulnerabilities/attacks) and LO2 (evaluate mitigation techniques)
Format	15 structured questions, no MCQs. Answer in your own words.
How to use this	Answer every question in the space provided. Together, your answers form a set of study notes for this topic. Use them directly when revising for the mid-semester test and final exam.

These questions are drawn directly from this week's lecture. Several ask you to evaluate a mitigation, not just describe an attack. That's deliberate: LO2 is specifically about weighing mitigation options, not just naming them. Write full answers, not one-line definitions.

Layer 4 Fundamentals

Q1. List Layer 4's mandatory tasks, common to both TCP and UDP, and then list what TCP adds on top of them. For each item TCP adds, explain in one sentence why UDP deliberately leaves it out. *[LO1]*

Q2. Explain what a socket is, and why the combination of source IP, source port, destination IP, and destination port matters more than the IP address alone. *[LO1]*

TCP Connection Establishment and Teardown

Q3. Describe the three-way handshake, and explain exactly what resource the server commits, and at which step, before it has any proof the client is genuine. Why does this detail matter for a later part of this worksheet? *[LO1]*

Q4. Describe how a TCP connection is torn down, and explain what the TIME_WAIT state is holding onto and why that could matter operationally, not just academically. *[LO1]*

TCP-Based Attacks

Q5. Explain how a SYN scan differs from a full connect scan, and why an attacker might prefer one over the other. *[LO1]*

Q6. Explain why FIN, Xmas, and NULL scans can sometimes get past a firewall that a normal SYN scan would be caught by. *[LO1]*

Q7. Explain what OS fingerprinting actually identifies, and why simply knowing a target's operating system is valuable to an attacker even before any vulnerability is confirmed. *[LO1]*

Q8. TCP Hijacking and TCP RST Injection exploit almost exactly the same vulnerability. Explain what that shared vulnerability is, and then explain what actually distinguishes the two attacks from each other in terms of what the attacker is trying to achieve. *[LO1]*

Q9. Using what you learned about resource reservation during the handshake, explain step by step how a SYN flood actually exhausts a server, and why spoofed source addresses make the attack more effective rather than less. *[LO1]*

Q10. Evaluate SYN cookies as a mitigation. Why is this described as removing the vulnerability rather than just tolerating it, compared to simply increasing the backlog size? *[LO2]*

UDP Fundamentals and Attacks

Q11. Explain why UDP port scanning is inherently harder and less reliable for an attacker than TCP port scanning, referencing what a UDP response (or lack of one) actually tells the scanner. *[LO1]*

Q12. A network administrator argues that disabling ICMP entirely would stop most reconnaissance against their network. Evaluate this claim using at least two of today's attacks in your answer. *[LO2, LO3]*

Q13. Explain why UDP floods do not need to establish anything before they can start doing damage, and why this makes UDP flooding cheaper for an attacker than an equivalent TCP-based flood would be. *[LO1]*

Bringing It Together

Q14. Both TCP's weaknesses and UDP's weaknesses trace back to the same underlying design assumption. State that assumption in your own words, and explain how it shows up differently in a TCP attack versus a UDP attack of your choosing. *[LO1, LO2]*

Q15. A small business has almost no security budget this year. Using this week's mitigations, recommend three that could realistically be implemented at little to no cost, and justify why each one still meaningfully reduces risk despite costing so little. *[LO2, LO3]*

Turning This Into Exam Revision

Once you've answered every question, you have the beginnings of a proper study-notes set for this topic. To make it exam-ready:

1. Reread each answer and check it would make sense to you in three months, without the lecture slides open.
2. The resource-reservation question in the handshake section is the single most exam-relevant idea this week. If you can explain it clearly without notes, most of the SYN flood questions become easy.
3. Keep this worksheet with your notes for the mid-semester test (MCQ, LO1 and LO2) and final exam (MCQ + structured essay, LO1 through LO5). The evaluation-style questions above are close in format to the structured-essay section.